Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Calculates the bending and torsional stresses in a solid round shaft, with stress-concentration factors, and the factors of safety against yield. It covers static loading only, not fatigue or hollow shafts.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Bending stress, with its stress concentration | $\sigma = K_t \cdot 32M/(\pi d^3)$ |
| Torsional shear stress, with its stress concentration | $\tau = K_{ts} \cdot 16T/(\pi d^3)$ |
| Maximum shear stress (Tresca) | $\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$ |
| von Mises equivalent stress | $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$ |
| Factors of safety against yield, the lower governing | $n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$ |

Basis: Maximum-shear-stress and distortion-energy theories[1](§5-4 and §5-5, 10th ed.); Shaft design for stress[1](§7-4, 10th ed.).

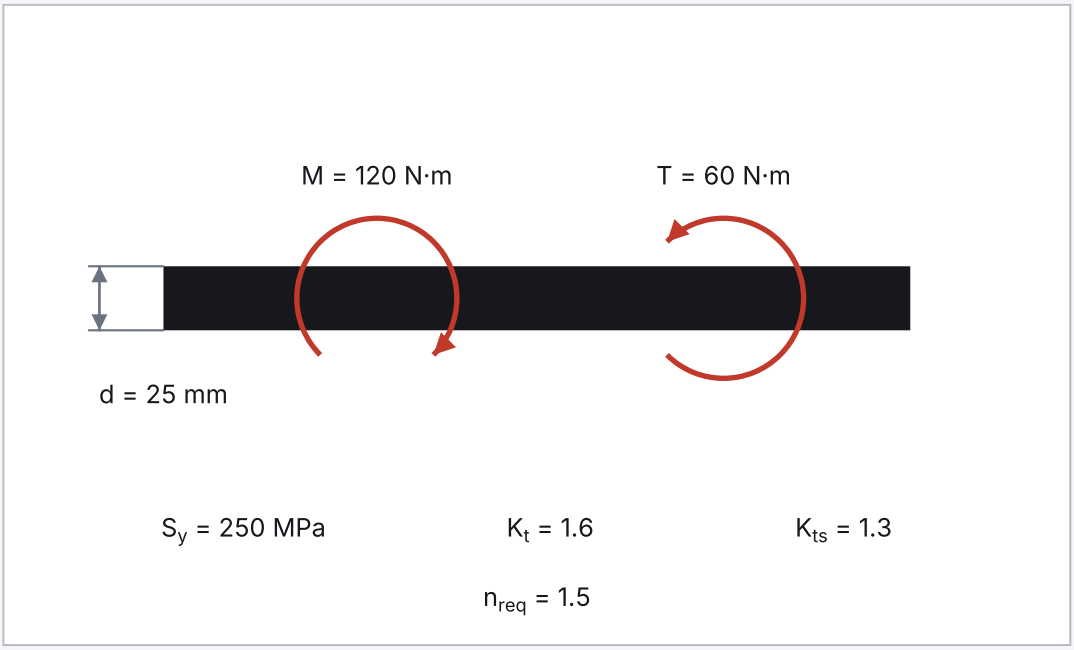



Figure 1 — A solid round shaft under bending and torsion at the section. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| d | Shaft diameter | 25 | mm | drawing: DRG-SH-402 rev A | 1 … 2000 mm |
| $S_y$ | Yield strength | 250 | MPa | drawing: DRG-SH-402 rev A | 50 … 3000 MPa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| M | Bending moment (magnitude) | 120 | N·m | spec: LOAD-SCHEDULE-01 | 0 … 10000000 N·m |
| T | Torque (magnitude) | 60 | N·m | spec: LOAD-SCHEDULE-01 | 0 … 10000000 N·m |
| $K_t$ | Bending stress-concentration factor (1 for none) | 1.6 | | spec: SPEC-MECH-03 design criteria | 1 … 5 |
| $K_{ts}$ | Torsional stress-concentration factor (1 for none) | 1.3 | | spec: SPEC-MECH-03 design criteria | 1 … 5 |
| $n_{req}$ | Required factor of safety | 1.5 | | spec: SPEC-MECH-03 design criteria | 1 … 10 |

## 3. Assumptions

A1. The shaft is a solid round section, linear elastic, under static loading. (The stress formulas of the method assume this.)

## 4. Working

[8] $\sigma = K_t * 32 * M / (pi * d^3)$ [1](§7-4)

$= 1.6 \times 32 \times 120\ N\cdot m / (\pi \times (25\ mm)^3)$

$= 125.16\ MPa$

Bending stress

[9] $\tau = K_{ts} * 16 * T / (pi * d^3)$ [1](§7-4)

$= 1.3 \times 16 \times 60\ N\cdot m / (\pi \times (25\ mm)^3)$

$= 25.424\ MPa$

Torsional shear stress

[10] $\tau_{max} = sqrt((\sigma / 2)^2 + \tau^2)$ [1](§5-4)

$= sqrt((125.16\ MPa / 2)^2 + (25.424\ MPa)^2)$

$= 67.549\ MPa$

Maximum shear stress (Tresca)

[11] $\sigma_{vm} = sqrt(\sigma^2 + 3 * \tau^2)$ [1](§5-5)

$= sqrt((125.16\ MPa)^2 + 3 \times (25.424\ MPa)^2)$

$= 132.68\ MPa$

von Mises stress

[12] $n_T = (S_y / 2) / \tau_{max}$

$= 250\ MPa / 2 / 67.549\ MPa$

$= 1.8505$

Factor of safety, maximum shear stress

[13] $n_{VM} = S_y / \sigma_{vm}$

$= 250\ MPa / 132.68\ MPa$

$= 1.8842$

Factor of safety, distortion energy

[14] $n_{min} = min(n_T, n_{VM})$

$= min(1.8505, 1.8842)$

$= 1.8505$

Governing factor of safety

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\tau_{max}$ | 67.55 MPa | Maximum shear stress |
| $\sigma_{vm}$ | 132.7 MPa | von Mises stress |
| $n_{min}$ | 1.8504 | Governing factor of safety |
| $\sigma$ | 125.2 MPa | Bending stress |
| $\tau$ | 25.42 MPa | Torsional shear |
| $n_T$ | 1.8504 | Factor of safety (Tresca) |
| $n_{VM}$ | 1.8841 | Factor of safety (von Mises) |
| $u_n$ | 0.81 | Utilisation of factor of safety against the required factor (governing) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Factor of safety against the required factor | 1.5 | 1.8505 | 0.81 | [1](§7-4) |

## 7. Applicability

None.

## 8. Verification

Template mech.shaft-stress@1.1.1 Content hash e68a699ed74d32f0 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.